

Template

Studentnames and studentnumbers here

2026-06-16

Set-up your environment

```
library(tidyverse)
library(readr)
library(ggplot2)
library(viridis)
library(sf)
```

Title Page

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Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Education and Income Inequality in the Netherlands Income inequality is an important societal concern in the Netherlands. Differences in income influence various aspects of life, such as level of living, healthcare, education, therefore research demonstrated how individual's place of residence and educational background can predetermine socioeconomic prospects for the future. As a result, people with similar abilities may experience different economic opportunities depending on their educational background and geographic location.

Key factor associated with income is level of education. Higher education may lead to higher prospects for employment and better wages. Regional exposure may also influence income through variations in labor market conditions, economic development, and access to educational resources. If income depends strongly on education level and region, policymakers can develop targeted measures to reduce inequalities and promote equal opportunities across different parts of the country.

In this report we quantify the relationship between level of education, region and income in the Netherlands using data from Statistics Netherlands (CBS). Specifically, we analyze how income varies across different education levels and regions and assess whether these factors are associated with economic outcomes.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
#onderwijs <- cbs_get_data("85525NED")
#inkomen <- cbs_get_data("86162NED")

#write_csv(onderwijs, "data/onderwijs.csv")
#write_csv(inkomen, "data/inkomen.csv")

data.inkomen <- read_csv("data/inkomen.csv")
data.onderwijs <- read_csv("data/onderwijs.csv")
```

2.2 Short summary of the dataset(s)

```
head(data.onderwijs)
```

```
##   Geslacht Leeftijd RegioS Perioden Bevolking_1 k_1BasisonderwijsVmboMbo1_2 k_2HavoVwoMbo24_3 k_3HboWoBachelor_8 k_32HboWoMasterDoctor_17
## 1  T001038   52052 NL01   2013JJ00   12724000                                31.8                                41.5                                17.3
## 2  T001038   52052 NL01   2014JJ00   12782300                                31.4                                41.4                                17.5
## 3  T001038   52052 NL01   2015JJ00   12849400                                30.8                                41.4                                17.8
## 4  T001038   52052 NL01   2016JJ00   12939500                                30.3                                41.6                                17.9
## 5  T001038   52052 NL01   2017JJ00   13028400                                29.6                                41.7                                18.3
## 6  T001038   52052 NL01   2018JJ00   13115700                                28.9                                41.5                                19.0
##   k_11Basisonderwijs_5 k_12VmboHavoVwoOnderbouwMbo1_6 k_21HavoVwoMbo24_7 k_31HboWoBachelor_8 k_32HboWoMasterDoctor_17
## 1                                10.3                                21.5                                41.5                                17.3
## 2                                10.3                                21.1                                41.4                                17.5
## 3                                10.2                                20.6                                41.4                                17.8
## 4                                10.1                                20.2                                41.6                                17.9
## 5                                9.7                                 19.9                                41.7                                18.3
## 6                                9.4                                 19.5                                41.5                                19.0
##   k_111Basisonderwijs_10 k_121VmboBKMbo1_11 k_122VmboGTHavoVwoOnderbouw_12 k_211Mbo2EnMbo3_13 k_212Mbo3_14 k_213HavoVwo_15 k_311HboWoBachelor_16 k_321HboWoMasterDoctor_17
## 1                                10.3                                13.5                                8.0                                17.4
## 2                                10.3                                13.0                                8.0                                17.3
## 3                                10.2                                12.6                                8.0                                17.2
## 4                                10.1                                12.3                                7.9                                17.2
## 5                                9.7                                 12.0                                7.9                                17.0
## 6                                9.4                                 11.7                                7.8                                16.8
##   k_213HavoVwo_15 k_311HboWoBachelor_16 k_321HboWoMasterDoctor_17
## 1                                9.4                                17.3                                9.5
## 2                                9.4                                17.5                                9.8
## 3                                9.4                                17.8                                10.0
## 4                                9.4                                17.9                                10.2
## 5                                9.5                                18.3                                10.3
## 6                                9.4                                19.0                                10.6
```

Dataset 1: Educational Attainment (85525NED)

The dataset “Bevolking; hoogstbehaald onderwijsniveau en regio” contains information on the highest educational attainment of people aged 15–75 in the Netherlands, broken down by region, gender, age group, and year. The data are collected by Statistics Netherlands (CBS) and are available from 2013–2024. The unit of observation is a combination of region and year.

Key variables include: - the percentage of people with low education (primary education, VMBO, MBO1), - medium education (HAVO, VWO, MBO2-4), and higher education (HBO, WO).

For this project, the share of highly educated people is the most relevant variable, as it can be compared with regional income levels. A limitation of the dataset is that it contains aggregated regional statistics rather than individual-level data, so relationships observed across regions may not necessarily apply to individuals.

```
head(data.inkomen)
```

```
##   Geslacht Persoonskenmerken RegioS Perioden Personen_1 PersonenMetPersoonlijkInkomen_2 AandeelPersoonlijkInkomen_3
## 1  T001038                T009002 NL01  2011JJ00      16309.7                    12807.2
## 2  T001038                T009002 NL01  2012JJ00      16378.0                    12891.6
## 3  T001038                T009002 NL01  2013JJ00      16429.5                    12951.4
## 4  T001038                T009002 NL01  2014JJ00      16454.5                    13005.8
## 5  T001038                T009002 NL01  2015JJ00      16528.1                    13115.5
## 6  T001038                T009002 NL01  2016JJ00      16595.2                    13225.4
##   GemiddeldGestandaardiseerdInkomen_4 MediaanGestandaardiseerdInkomen_5 GemiddeldPersoonlijkInkomen_6
## 1                                26.9                                24.2                                28.4
## 2                                27.1                                24.4                                29.4
## 3                                27.3                                24.6                                29.4
## 4                                28.9                                25.4                                29.4
## 5                                28.7                                25.9                                30.4
## 6                                30.0                                27.0                                31.4
##   MediaanPersoonlijkInkomen_7
## 1                                23.1
## 2                                23.1
## 3                                23.1
## 4                                23.3
## 5                                23.5
## 6                                24.0
```

Dataset 2: Personal Income (86162NED)

The dataset “Inkomen van personen; persoonskenmerken, regio” (CBS) contains information on personal income by region and demographic characteristics in the Netherlands. The unit of observation is a combination of region, year, gender, and personal characteristics.

Key variables include: - RegioS: region code - Perioden: year - GemiddeldPersoonlijkInkomen: average personal income (1,000 euros) - MediaanPersoonlijkInkomen: median personal income (1,000 euros)

The dataset covers the period 2011–2024. Figures for 2024 are provisional. A limitation of the dataset is that it contains aggregated regional data rather than individual-level observations. For our analysis, the most relevant variable is median personal income or average personal income, because these can be compared with regional educational attainment. A limitation is that the dataset is aggregated by region, so it does not allow conclusions about individual people. Also, the 2024 figures are provisional, and regional averages may hide differences within municipalities or provinces.

These datasets can be merged using the region (RegioS) and year (Perioden) variables, allowing us to examine the relationship between educational attainment and personal income across Dutch regions.

Dataset	Unit of observation	Key variables	Source
Onderwijs (85525NED)	One Dutch region in one year	Region, year, population size, share low education, share medium education, share high education	Statistics Netherlands (CBS StatLine), 2013–2024
Inkomen (86162NED)	One Dutch region in one year	Region, year, average personal income, median personal income, number of persons with income	Statistics Netherlands (CBS StatLine), 2011–2024

```
inline_code = TRUE
```

2.3 Describe the type of variables included

Educational Attainment Dataset (data.onderwijs) (85525NED)

This dataset primarily contains socio-economic status (SES) information, as educational attainment is a common indicator of an individual's socio-economic position. The dataset includes variables on the highest completed level of education, population counts, age groups, gender, region, and year.

The data are obtained from administrative sources and registers maintained by Statistics Netherlands (CBS). Educational attainment is compiled using information from educational institutions and government records. The dataset is aggregated at the regional level, meaning that it reports percentages and population counts for groups rather than information on individual persons.

Income of Persons Dataset (data.inkomen) (86162NED)

The variables in this dataset mainly contain socio-economic status (SES) information, as they measure personal income levels across different regions and population groups. Key variables include average personal income, median personal income, the number of persons with income, region, year, gender, and other demographic characteristics.

The data are derived from administrative sources, such as tax records and social security registers, and are compiled by Statistics Netherlands (CBS). The information is therefore not collected through surveys or interviews. The dataset is aggregated by region and demographic group, meaning that it provides summary statistics rather than individual-level income data.

Part 3 - Quantifying

3.1 Data cleaning

```
colSums(is.na(data.inkomen))
```

```
##                Geslacht                Persoonskenmerken
##                0                0
##                Perioden                Personen_1        PersonenMetPersoonlijkInk
##                0                0
##                AandeelPersonen_3  GemiddeldGestandaardiseerdInkomen_4  MediaanGestandaardiseerdInk
##                0                11057
##                GemiddeldPersoonlijkInkomen_6        MediaanPersoonlijkInkomen_7
##                32308                32231
```

```
colSums(is.na(data.onderwijs))
```

```
##                Geslacht                Leeftijd                RegioS
##                0                1                0
##                Perioden                Bevolking_1        k_1BasisonderwijsVmboMbo1_2
##                0                9314                13503
##                k_2HavoVwoMbo24_3                k_3HboWo_4        k_11Basisonderwijs_5
##                13503                13503                13503
##                k_12VmboHavoVwoOnderbouwMbo1_6        k_21HavoVwoMbo24_7        k_31HboWoBachelor_8
##                13503                13503                13503
##                k_32HboWoMasterDoctor_9        k_111Basisonderwijs_10        k_121VmboBKMbo1_11
##                13503                13503                13503
##                k_122VmboGTHavoVwoOnderbouw_12        k_211Mbo2EnMbo3_13        k_212Mbo4_14
##                13503                13503                13503
##                k_213HavoVwo_15        k_311HboWoBachelor_16        k_321HboWoMasterDoctor_17
##                13503                13503                13503
```

```
#check for missing values
```

```
sum(duplicated(data.inkomen))
```

```
## [1] 0
```

```
sum(duplicated(data.onderwijs))
```

```
## [1] 7
```

```
#check for duplicates
```

```
str(data.inkomen)
```

```
## 'data.frame':    419244 obs. of  11 variables:
## $ Geslacht                : chr  "T001038" "T001038" "T001038" "T001038" ...
## $ Persoonskenmerken        : chr  "T009002" "T009002" "T009002" "T009002" ...
## $ RegioS                   : chr  "NL01  " "NL01  " "NL01  " "NL01  " ...
## $ Perioden                 : chr  "2011JJ00" "2012JJ00" "2013JJ00" "2014JJ00" ...
## $ Personen_1               : num  16310 16378 16430 16454 16528 ...
## $ PersonenMetPersoonlijkInkomen_2 : num  12807 12892 12951 13006 13116 ...
## $ AandeelPersonen_3        : int   100 100 100 100 100 100 100 100 100 100 ...
## $ GemiddeldGestandaardiseerdInkomen_4: num  26.9 27.1 27.3 28.9 28.7 30 31.1 31.5 33.9 34.5 ...
## $ MediaanGestandaardiseerdInkomen_5 : num  24.2 24.4 24.6 25.4 25.9 27 27.8 28.2 29.7 31 ...
## $ GemiddeldPersoonlijkInkomen_6      : num  28.8 29.1 29.4 29.9 30.4 31 32 32.3 33.4 34.3 ...
## $ MediaanPersoonlijkInkomen_7        : num  23.1 23.1 23.1 23.3 23.5 24 24.6 25.3 26.3 27.2 ...
```

```
str(data.onderwijs)
```

```
## 'data.frame': 70585 obs. of 21 variables:
## $ Geslacht : chr "T001038" "T001038" "T001038" "T001038" ...
## $ Leeftijd : int 52052 52052 52052 52052 52052 52052 52052 52052 52052 52052
## $ RegioS : chr "NL01 " "NL01 " "NL01 " "NL01 " ...
## $ Perioden : chr "2013JJ00" "2014JJ00" "2015JJ00" "2016JJ00" ...
## $ Bevolking_1 : int 12724000 12782300 12849400 12939500 13028400 13115700 132061
## $ k_1BasisonderwijsVmboMbo1_2 : num 31.8 31.4 30.8 30.3 29.6 28.9 28.3 27.4 26.5 26.3 ...
## $ k_2HavoVwoMbo24_3 : num 41.5 41.4 41.4 41.6 41.7 41.5 41.4 41.5 41.8 41.4 ...
## $ k_3HboWo_4 : num 26.7 27.3 27.8 28.1 28.7 29.6 30.3 31.1 31.7 32.3 ...
## $ k_11Basisonderwijs_5 : num 10.3 10.3 10.2 10.1 9.7 9.4 9.4 9 8.6 8.7 ...
## $ k_12VmboHavoVwoOnderbouwMbo1_6 : num 21.5 21.1 20.6 20.2 19.9 19.5 18.9 18.4 18 17.6 ...
## $ k_21HavoVwoMbo24_7 : num 41.5 41.4 41.4 41.6 41.7 41.5 41.4 41.5 41.8 41.4 ...
## $ k_31HboWoBachelor_8 : num 17.3 17.5 17.8 17.9 18.3 19 19.3 19.7 19.9 20.2 ...
## $ k_32HboWoMasterDoctor_9 : num 9.5 9.8 10 10.2 10.3 10.6 10.9 11.3 11.8 12.1 ...
## $ k_111Basisonderwijs_10 : num 10.3 10.3 10.2 10.1 9.7 9.4 9.4 9 8.6 8.7 ...
## $ k_121VmboBKMbo1_11 : num 13.5 13 12.6 12.3 12 11.7 11.1 10.8 10.3 10.2 ...
## $ k_122VmboGTHavoVwoOnderbouw_12 : num 8 8 8 7.9 7.9 7.8 7.8 7.6 7.6 7.4 ...
## $ k_211Mbo2EnMbo3_13 : num 17.4 17.3 17.2 17.2 17 16.8 16.7 16.6 16.4 16.2 ...
## $ k_212Mbo4_14 : num 14.6 14.7 14.8 15 15.2 15.3 15.3 15.5 15.8 15.7 ...
## $ k_213HavoVwo_15 : num 9.4 9.4 9.4 9.4 9.5 9.4 9.4 9.5 9.5 9.5 ...
## $ k_311HboWoBachelor_16 : num 17.3 17.5 17.8 17.9 18.3 19 19.3 19.7 19.9 20.2 ...
## $ k_321HboWoMasterDoctor_17 : num 9.5 9.8 10 10.2 10.3 10.6 10.9 11.3 11.8 12.1 ...
```

```
data.inkomen <- na.omit(data.inkomen)
data.onderwijs <- na.omit(data.onderwijs)
#remove no data
```

Filtering for data.inkomen

```
data.inkomenclean <- data.inkomen %>%
  mutate(
    Geslacht = trimws(as.character(Geslacht)),
    Persoonskenmerken = trimws(as.character(Persoonskenmerken)),
    RegioS = trimws(as.character(RegioS)),
    Perioden = trimws(as.character(Perioden))
  ) %>%
  filter(
    Geslacht == "T001038",
    Persoonskenmerken == "53050",
    grepl("^LD", RegioS) |
    grepl("^PV", RegioS) |
    RegioS == "GM0363"
  ) %>%
  select(
    RegioS,
    Perioden,
    GemiddeldPersoonlijkInkomen_6,
  )

head(data.inkomenclean)
```

```
##   RegioS Perioden GemiddeldPersoonlijkInkomen_6
## 1   LD01 2011JJ00                      9.7
## 2   LD01 2012JJ00                      9.5
## 3   LD01 2013JJ00                      9.3
## 4   LD01 2014JJ00                      9.2
## 5   LD01 2015JJ00                      9.3
## 6   LD01 2016JJ00                      9.5
```

Chosen variables: RegioS - region identifier Perioden - time period GemiddeldPersoonlijkInkomen_6 - mean personal income (average)

Filtering for data.onderwijs

```
data.onderwijsclean <- data.onderwijs %>%
  filter(
    trimws(Geslacht) == "T001038",
    Leeftijd %in% c(53050, 53700)
  ) %>%
  mutate(
    RegioS = trimws(RegioS),
    Perioden = trimws(Perioden)
  ) %>%
  group_by(RegioS, Perioden) %>%
  summarise(
    k_2HavoVwoMbo24_3 = mean(k_2HavoVwoMbo24_3, na.rm = TRUE),
    k_3HboWo_4 = mean(k_3HboWo_4, na.rm = TRUE),
    k_32HboWoMasterDoctor_9 = mean(k_32HboWoMasterDoctor_9, na.rm = TRUE),
    .groups = "drop"
  )
```

Chosen variables: RegioS - region identifier Perioden - time period k_2HavoVwoMbo24_3 - Medium education variable k_3HboWo_4 - Higher education variable k_32HboWoMasterDoctor_9 - Bachelor + Master/Doctor variable (Filtered with 53050 and 53700 age groups and took an average to not create identical tables)

Merging two data sets

```
data.combined <- data.inkomenclean %>%
  inner_join(data.onderwijsclean, by = c("RegioS", "Perioden"))

nrow(data.combined)
```

```
## [1] 204
```

```
ncol(data.combined)
```

```
## [1] 6
```

```
head(data.combined)
```

```
##   RegioS Perioden GemiddeldPersoonlijkInkomen_6 k_2HavoVwoMbo24_3 k_3HboWo_4 k_32HboWoMasterDoctor_9
## 1   LD01 2013JJ00                      9.3                48.70        18.75                5.00
```

## 2	LD01	2014JJ00	9.2	48.35	19.40	5.15
## 3	LD01	2015JJ00	9.3	48.20	19.85	5.25
## 4	LD01	2016JJ00	9.5	48.20	19.80	5.40
## 5	LD01	2017JJ00	9.8	47.35	20.55	5.70
## 6	LD01	2018JJ00	10.3	47.25	21.40	5.95

3.2 Generate necessary variables

1. variable: mid education share
2. variable: high education share
3. variable: income index

```
data.combined <- data.combined %>%
  mutate(
    high_education_share = k_3HboWo_4 + k_32HboWoMasterDoctor_9,
    mid_education_share  = k_2HavoVwoMbo24_3,
    income_index         = GemiddeldPersoonlijkInkomen_6 /
                          mean(GemiddeldPersoonlijkInkomen_6, na.rm = TRUE) * 100
  )
```

3.3 Temporal variation — average income index and education share over time

```
data.combined <- data.combined %>%
  mutate(year = as.numeric(substr(Perioden, 1, 4)))
```

```
temporal_data <- data.combined %>%
  group_by(year) %>%
  summarise(mean_income = mean(GemiddeldPersoonlijkInkomen_6, na.rm = TRUE),
            mean_high_edu_share = mean(high_education_share, na.rm = TRUE)
  )
```

```
ggplot(temporal_data, aes(x = year)) +
  geom_line(aes(y = mean_income, colour = "Average income (×1000 EUR)",
                linewidth = 0.8)) +
  geom_point(aes(y = mean_income, colour = "Average income (×1000 EUR)",
                 size = 2)) +
  geom_line(aes(y = mean_high_edu_share, colour = "Higher education share (%)",
                linewidth = 0.8, linetype = "dashed")) +
  geom_point(aes(y = mean_high_edu_share, colour = "Higher education share (%)",
                 size = 2)) +
  geom_vline(xintercept = 2020, linetype = "dotted", colour = "grey40") +
  annotate("text", x = 2020.1, y = 36, label = "COVID-19",
          hjust = 0, size = 3, colour = "grey40") +
  scale_x_continuous(breaks = seq(min(temporal_data$year),
                                  max(temporal_data$year), by = 1)) +
  scale_colour_manual(values = c("Average income (×1000 EUR)" = "dodgerblue",
                                "Higher education share (%)" = "tomato")) +
  labs(
    title = "Average income and higher education share across Dutch provinces, 2013-2024",
    x     = "Year",
```

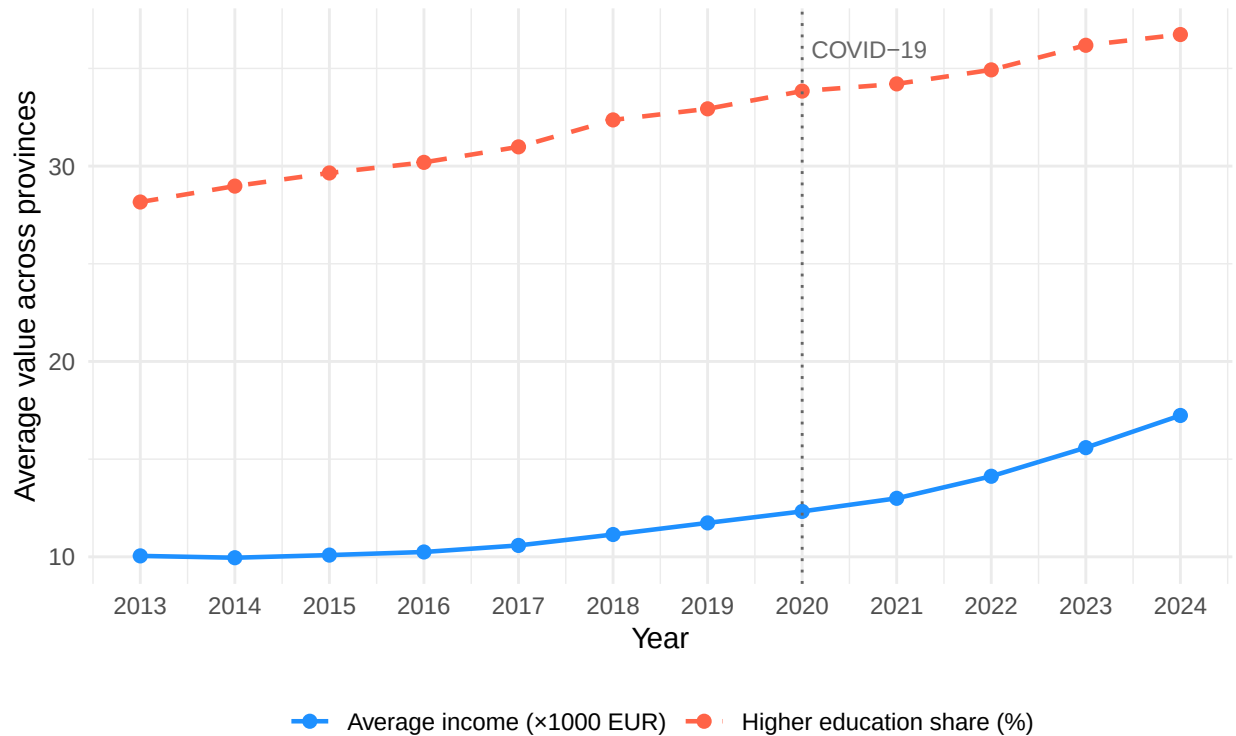


```

y      = "Average value across provinces",
colour = NULL,
caption = "Source: Statistics Netherlands (CBS)."
) +
theme_minimal() +
theme(legend.position = "bottom")

```

Average income and higher education share across Dutch provinces, 2013–



Source: Statistics Netherlands (CBS).

Between 2013 and 2024, both average personal income and the share of higher-educated residents increased steadily across Dutch provinces. Average income rose from approximately €10,000 to €17,000, while the higher education share grew from 28% to 37%. The two trends move in parallel, suggesting a positive association between education levels and income over time. Growth in both indicators accelerated after 2018. The dotted line marks the onset of COVID-19 in 2020; while no sharp break is visible, income growth slowed slightly in 2019–2020 before recovering strongly. Note that rising income partly reflects inflation rather than real income gains, and the parallel trends do not establish a causal relationship — both may reflect broader urbanisation and economic growth.

```

map_data <- provinces_sf %>%
  left_join(
    data.provinces %>%
      left_join(prov_lookup, by = "RegioS") %>%
      group_by(statnaam) %>%
      summarise(
        mean_income = mean(GemiddeldPersoonlijkInkomen_6, na.rm = TRUE),
        mean_high_edu = mean(high_education_share, na.rm = TRUE)
      )
  ),

```

```

    by = "statnaam"
  )

```

3.4 Visualize spatial variation

```

library(ggmap)

ggplot(map_data) +
  geom_sf(aes(fill = mean_high_edu), colour = "white", linewidth = 0.5) +
  scale_fill_gradient(
    name = "Higher education\nshare (%)",
    low = "#deebf7",
    high = "#08306b"
  ) +
  geom_sf_text(aes(label = statnaam),
    size = 2.5,
    colour = "black",
    fontface = "bold") +
  geom_sf_text(data = map_data %>% filter(statnaam %in% c("Utrecht")),
    aes(label = statnaam),
    size = 2.5,
    colour = "white",
    fontface = "bold") +

  labs(
    title = "Average higher education share by Dutch province, 2013-2024",
    caption = "Source: Statistics Netherlands (CBS). Boundary: CBS/cartomap 2024."
  ) +
  theme_void() +
  theme(
    plot.title = element_text(size = 12, margin = margin(b = 8)),
    legend.position = "right"
  )

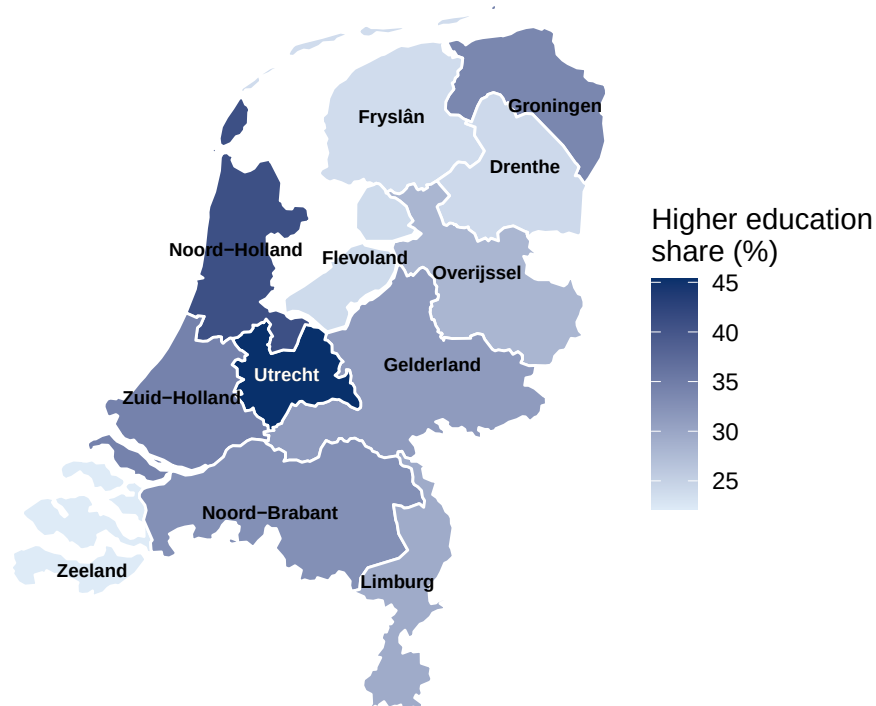
```

```

## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give correct results f
## data
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give correct results f
## data

```

Average higher education share by Dutch province, 2013–2024



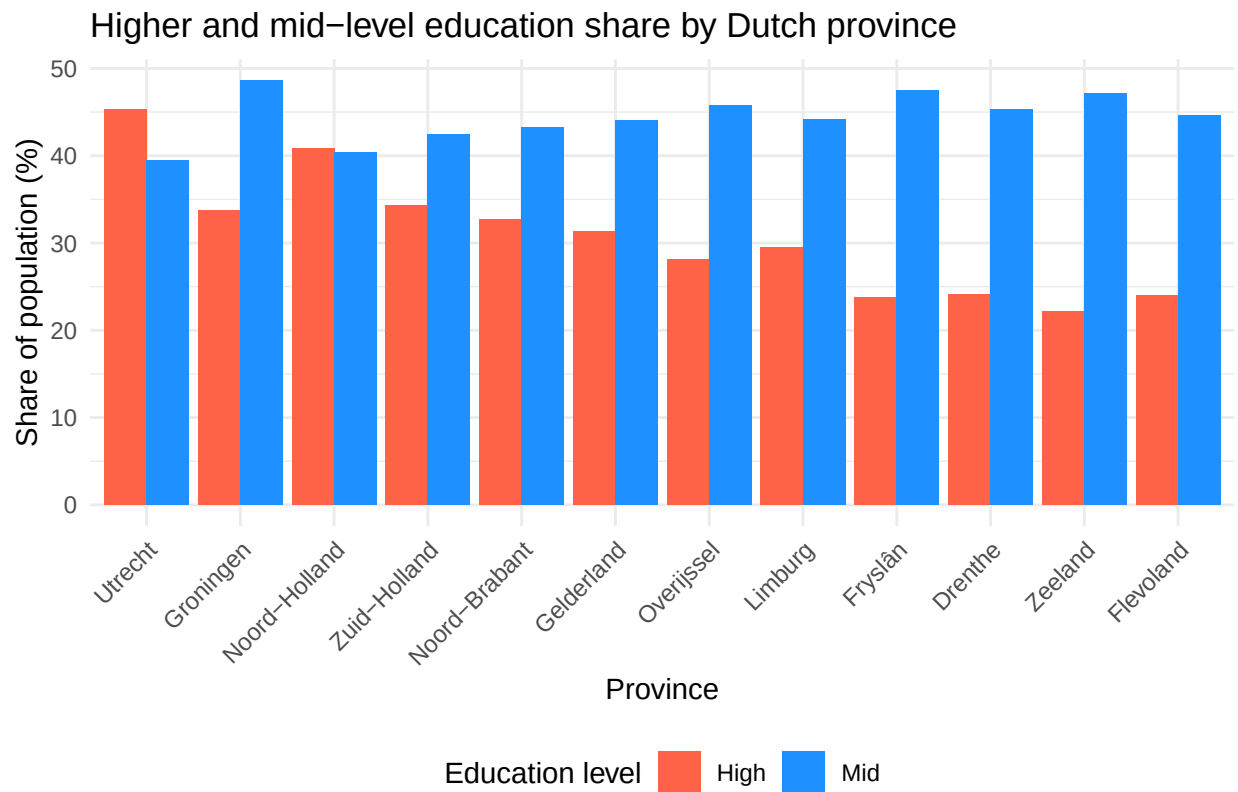
Source: Statistics Netherlands (CBS). Boundary: CBS/cartomap 2024.

The map reveals a clear urban-rural divide in higher education across Dutch provinces. Utrecht (~45%) and Noord-Holland (~41%) stand out as the most highly educated, reflecting the concentration of universities and knowledge-intensive industries in the Randstad. In contrast, peripheral provinces like Zeeland (~22%), Fryslân (~24%) and Drenthe (~24%) have the lowest shares. Groningen is a notable exception among northern provinces, likely due to its large university. Provinces with higher education shares generally also have higher average incomes, suggesting education partly drives regional income differences. However, Zeeland — the least educated province — records the highest average income (~€14k), indicating that education alone does not fully explain provincial income variation.

3.5 Visualize sub-population variation

```
# reshape to long format for plotting
subpop_data <- data.combined %>%
  filter(substr(RegioS, 1, 2) == "PV") %>%
  left_join(prov_lookup, by = "RegioS") %>%
  group_by(statnaam) %>%
  summarise(
    High = mean(high_education_share, na.rm = TRUE),
    Mid = mean(mid_education_share, na.rm = TRUE)
  ) %>%
  pivot_longer(cols = c(High, Mid),
               names_to = "education_level",
               values_to = "share")
```

```
ggplot(subpop_data, aes(x = reorder(statnaam, -share),
                          y = share,
                          fill = education_level)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("High" = "tomato",
                              "Mid" = "dodgerblue")) +
  labs(
    title = "Higher and mid-level education share by Dutch province",
    x = "Province",
    y = "Share of population (%)",
    fill = "Education level",
    caption = "Source: Statistics Netherlands (CBS)."
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
)
```



Source: Statistics Netherlands (CBS).

This grouped bar chart is directly relevant to understanding regional income inequality in the Netherlands. If higher education is a key driver of personal income — as suggested by the scatter plots in sections 3.5 and 3.6 — then provinces with low higher education shares are structurally disadvantaged in terms of earning potential. The large variation in higher education share across provinces (22% in Zeeland to 45% in Utrecht) implies that where someone is born or lives in the Netherlands significantly affects their access to high-income opportunities. Mid-level education, by contrast, is evenly distributed across provinces and therefore cannot explain regional income differences. This means that policies targeting higher education access in peripheral

provinces like Zeeland, Fryslân and Drenthe could be an effective lever for reducing the income gap between Dutch regions.

3.6 Event analysis

```
event_data <- data.combined %>%
  filter(substr(RegioS, 1, 2) == "PV") %>%
  group_by(year) %>%
  summarise(
    mean_income = mean(GemiddeldPersoonlijkInkomen_6, na.rm = TRUE),
    mean_high_edu = mean(high_education_share, na.rm = TRUE)
  ) %>%
  mutate(period = ifelse(year < 2020, "Pre-COVID", "Post-COVID"))

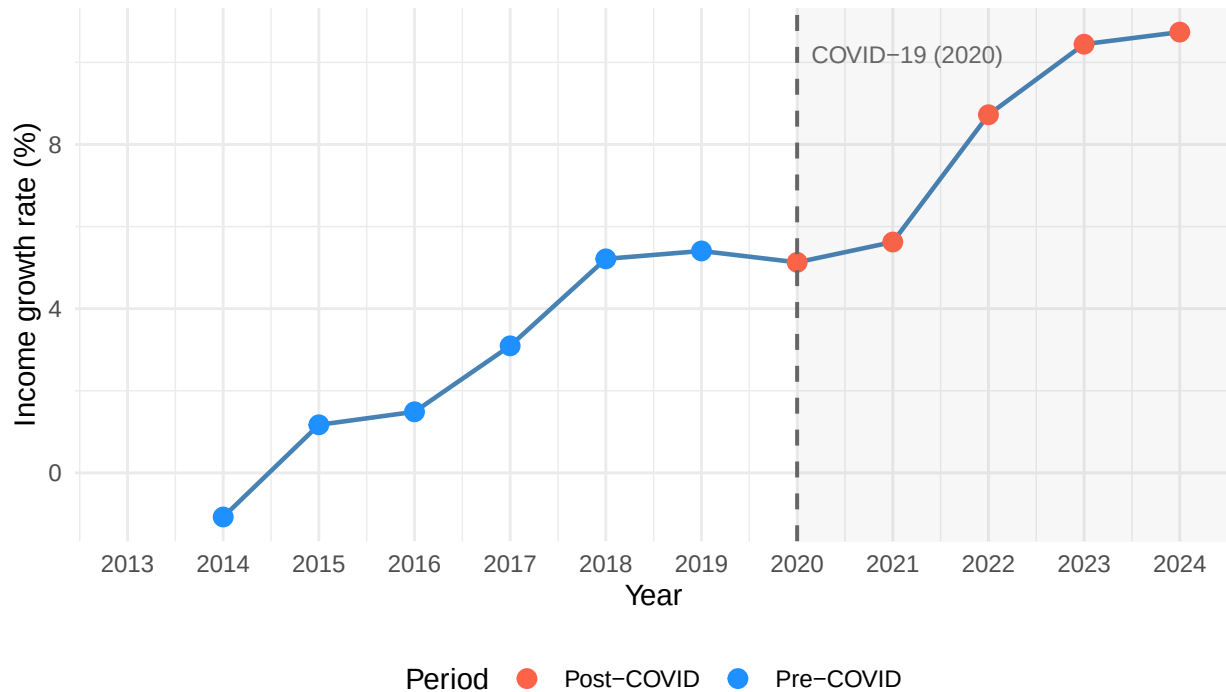
event_data_growth <- event_data %>%
  arrange(year) %>%
  mutate(income_growth = (mean_income - lag(mean_income)) /
         lag(mean_income) * 100)

ggplot(event_data_growth, aes(x = year, y = income_growth)) +
  geom_line(colour = "steelblue", linewidth = 0.8) +
  geom_point(aes(colour = period), size = 3) +
  geom_vline(xintercept = 2020, linetype = "dashed",
            colour = "grey40", linewidth = 0.7) +
  annotate("text", x = 2020.15, y = max(event_data_growth$income_growth, na.rm = TRUE) * 0.95,
         label = "COVID-19 (2020)", hjust = 0, size = 3, colour = "grey40") +
  annotate("rect", xmin = 2020, xmax = Inf,
         ymin = -Inf, ymax = Inf,
         alpha = 0.05, fill = "grey40") +
  scale_colour_manual(values = c("Pre-COVID" = "dodgerblue",
                                "Post-COVID" = "tomato")) +
  scale_x_continuous(breaks = seq(min(event_data_growth$year, na.rm = TRUE),
                                max(event_data_growth$year, na.rm = TRUE), by = 1)) +
  labs(
    title = "Annual growth rate of average personal income before and after COVID-19",
    subtitle = "Dutch provinces, 2013-2024",
    x = "Year",
    y = "Income growth rate (%)",
    colour = "Period",
    caption = "Source: Statistics Netherlands (CBS)."
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range ('geom_line()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range ('geom_point()').
```

Annual growth rate of average personal income before and after COVID-19 Dutch provinces, 2013–2024



Source: Statistics Netherlands (CBS).

Prior to COVID-19, income growth was modest and gradually increasing, averaging around 1–5% annually. Contrary to expectations, the onset of COVID-19 in 2020 did not reduce income growth — the Dutch government’s NOW-regeling wage subsidy scheme cushioned the impact on household incomes. More striking is the sharp post-COVID acceleration, with growth rates nearly doubling from ~5% in 2020 to ~10% by 2024, likely driven by inflation compensation and a tight labour market particularly benefiting higher-educated workers. Also to note that this is an association, not a causal claim — the upward trend in income growth began before 2020, suggesting broader structural factors beyond COVID are at play.

```
library(tidyverse)
library(gganimate)

bubble_data_years <- data.combined %>%
  filter(substr(RegioS, 1, 2) == "PV") %>%
  left_join(prov_lookup, by = "RegioS") %>%
  filter(year %in% c(2013, 2016, 2019, 2022, 2024))

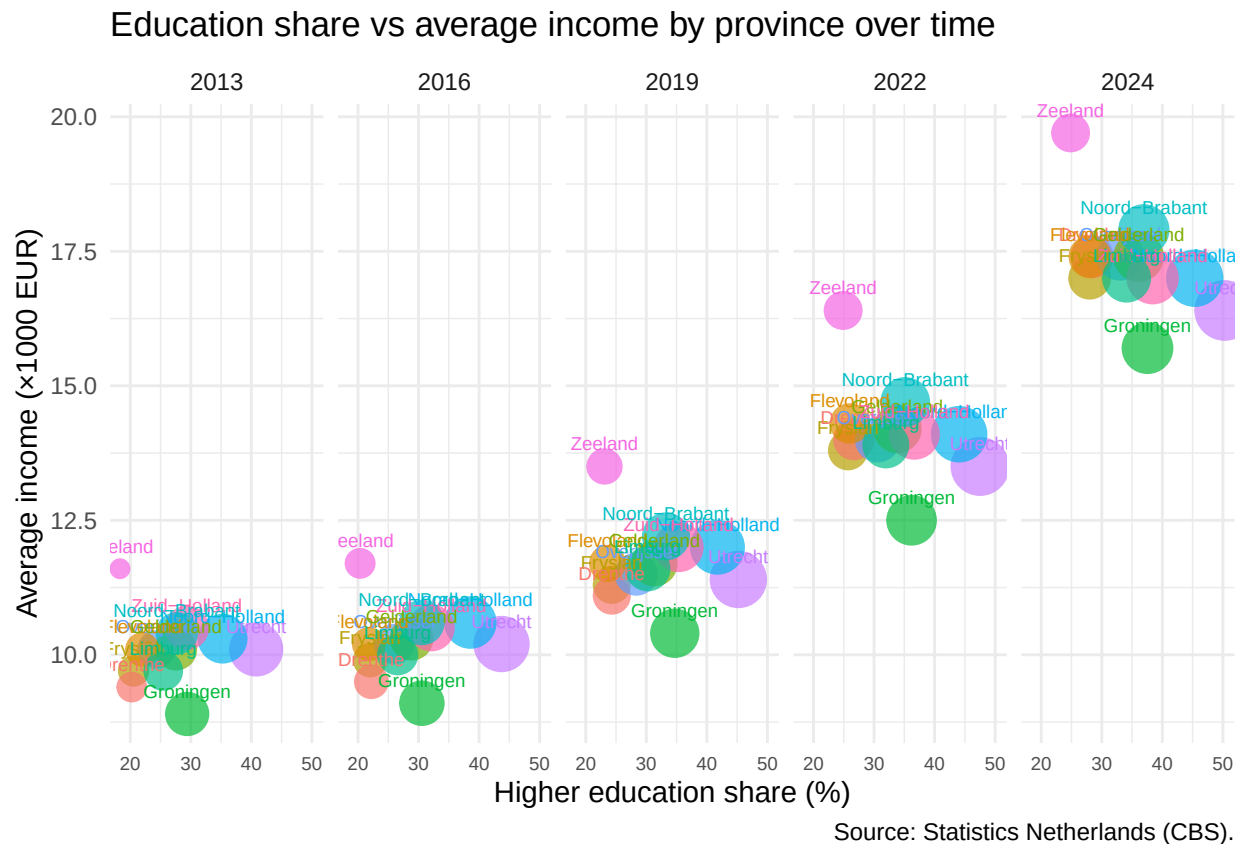
ggplot(bubble_data_years, aes(x = high_education_share,
                             y = GemiddeldPersoonlijkInkomen_6,
                             size = high_education_share,
                             colour = statnaam)) +

  geom_point(alpha = 0.7) +
  geom_text(aes(label = statnaam),
            size = 2.5, vjust = -1.2, show.legend = FALSE) +
  scale_size(range = c(3, 10), guide = "none") +
  facet_wrap(~year, nrow = 1) +
  labs(
    title = "Education share vs average income by province over time",
```

```

x      = "Higher education share (%)",
y      = "Average income (×1000 EUR)",
colour = "Province",
caption = "Source: Statistics Netherlands (CBS)."
) +
theme_minimal() +
theme(legend.position = "none",
      axis.text.x = element_text(size = 7))

```



““

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.